EasyPaper: A Lightweight Local Paper Reader

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Abstract

We present a compact desktop system for reading, translating and questioning academic papers. The system keeps the document structure, extracts text locally, runs OCR only for scanned pages, and sends chunked text to an OpenAI-compatible endpoint.

Reading a paper on a desktop should not require uploading the whole PDF to a remote service. EasyPaper therefore performs layout analysis and OCR in a separate local process, preserving the embedded text layer whenever it is available. The translation layer keeps headings, paragraphs, tables, formulas, captions and reading order.

Blocks are grouped into batches of roughly 3,600 characters. Every request returns a mapping from stable block ids to translations, so the display order cannot drift even when requests complete out of order. Formula blocks are preserved verbatim and table cells keep their Markdown grid.

The PDF is parsed into blocks with PyMuPDF. A two-column reading-order heuristic sorts each band before sending blocks to the translator. The OCR worker is lazy: it is started only when a page has little or no text and a large image area.

```
E = m c^2 + sum_i x_i
```

The equation above is a structural placeholder used to check that formula blocks are not reflowed into prose. Captions and tables are rendered as separate block types.

5 Results and Discussion

Table 1 summarizes the latency of each stage. Local parsing dominates startup, while OCR is only paid on scanned documents. The interface remains responsive because every long-running task executes outside the GUI thread.

Stage	Latency	Notes
Parse	0.4 s	local
OCR	8.2 s / page	scanned only
Translate	API-bound	chunked

Table 1: Stage latency breakdown.

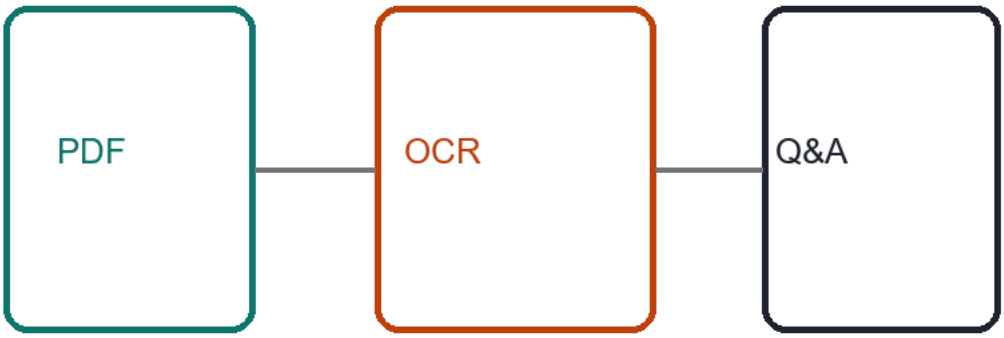

Figure 1: Local OCR pipeline.

References

[1] A. Example. Local-first document tools. In Proceedings of ExampleConf, 2026.
[2] B. Sample. Practical PDF layout analysis. Journal of Examples, 12(3):1-10, 2025.

6 Scanned Appendix

This page intentionally contains only a raster image. A local OCR model must be used before translation.

Step	Action	Output
1	Render page	RGB image